

Solar Panel Power Prediction in Aswan, Egypt

Abstract

Knowing the power output of solar panels (Photovoltaics) in advance is essential for regulating energy consumption. While several studies have historically predicted solar panel power, recent climate change in Egypt has transformed this into a new and complex problem. This project addresses the challenge of predicting PV output in Aswan, Egypt, a region significantly impacted by these atmospheric shifts. Utilizing a dataset of meteorological measurements spanning 2021–2022, we explored the relationship between environmental factors and solar efficiency. Through a multi-stage process involving automated data diagnostics, rigorous exploratory data analysis (EDA), and machine learning refinement, we established an implemented process to predict power output based on recent features. This report details the discovery phase, pre-processing strategies, and statistical validation, providing a foundation for high-precision energy forecasting in Egypt's changing climate.

1. Introduction

The regulation of energy consumption relies heavily on the ability to forecast solar power generation accurately. In Egypt, climate shifts have rendered traditional prediction models less effective, necessitating a fresh approach to the problem. This project aims to establish an implemented process to predict the power output of solar panels located in Egypt based on features captured in recent years.

The project encompasses the full data lifecycle as required by the assignment:

- **Analysis and Discovery:** Understanding the underlying drivers of power variance through statistical exploration.
- **Pre-processing:** Cleaning and transforming raw data into predictive features, addressing issues like multicollinearity and memory optimization.
- **Modeling:** Implementing and comparing various machine learning architectures, including Gradient Boosting and Deep Learning.
- **Interpretation and Validation:** Statistically verifying the model's logic and performance on unseen data.

Techniques Used:

- Automated Data Profiling (Custom Diagnostic Tool)
- Hypothesis Testing (ANOVA, Chi-Square)
- Temporal & Cyclical Feature Engineering
- Ensemble Learning (XGBoost, ExtraTrees)
- Deep Learning (Feed-Forward Neural Networks, RNN/LSTM)

2. Methodology

The project follows a standard data science lifecycle:

1. **Data Discovery:** Initial profiling using a custom diagnostic tool to identify multicollinearity and data quality issues.
2. **EDA:** Statistical validation of feature importance through ANOVA and Chi-Square tests to ensure features are physically and statistically grounded.
3. **Preprocessing:** Feature engineering of seasonal, cyclical, and lagged variables to capture the time-dependent nature of solar irradiation.
4. **Modeling:** Comparative evaluation of regression and classification approaches across different feature subsets.
5. **Optimization:** Hyperparameter tuning of ensemble models and sequence modeling with Deep Learning.

3. Proposed Model

The proposed architecture utilizes an **Ensemble Regression Pipeline** alongside **Deep Learning experiments**:

- **Phase 1 (Preprocessing):** Missing value verification, One-Hot Encoding for categorical data, and Standard Scaling for numerical weather features.
- **Phase 2 (Feature Engineering):** Extraction of DayOfYear, Month, and Season, alongside 1-day and 7-day power lags.
- **Phase 3 (Ensemble Modeling):** Implementation of tree-based regressors (ExtraTrees/XGBoost) to capture non-linear interactions.
- **Phase 4 (Deep Learning):** Implementation of **FFRegressor** (Multilayer Perceptron) and **RNNRegressor** (SimpleRNN) to test sequential dependencies.

4. Results and Discussion

4.1 Dataset Description

The dataset, AswanData_weatherdata.csv, contains 398 daily observations from April 2021 to April 2022.

- **Features:** AvgTemperature, Humidity, Wind, Pressure, AverageDew.
- **Target:** Solar(PV) (Range: 8.58 – 40.04 units).

4.2 Preprocessing & Data Analysis

- **Data Profiling (Custom Tool):** Flagged severe multicollinearity between Temperature and Humidity.
- **ANOVA Results:** *Solar(PV)* vs *Season* yielded $F\text{-stat} = 32.68$ ($p < 0.001$), confirming seasonality as the dominant driver.
- **Chi-Square Test:** *Season* vs *PressureBin* ($p \approx 2.08 \times 10^{-48}$) indicated a

strong dependency between these contextual variables.

4.3 Modeling Results (ExtraTrees)

The final optimized model transitioned back to an ensemble approach to achieve higher precision. with the feature set: ('DayOfYear', 'AvgTemperture', 'Wind', 'Solar_lag1', 'Solar_lag7', 'Season', 'PressureBin').

Metric	ExtraTrees Result
Test R^2	0.8656
Test MAE	1.7807
Test RMSE	2.6628

5. Interpretation and Validation

The validation phase highlights a clear hierarchy in model performance. While the **FFRegressor** achieved a respectable validation RMSE of **4.16**, the **ExtraTrees** model significantly outperformed all architectures with an RMSE of **2.66**.

The **RNNRegressor's** poor performance (RMSE 8.00) suggests that the daily volatility in the Aswan dataset—likely due to the "new problem" of climate-induced weather anomalies—introduces too much noise for a SimpleRNN to extract meaningful 7-day sequences. In contrast, the **ExtraTrees** model successfully used the engineered **Season** and **DayOfYear** features to navigate this volatility.

6. Conclusion and Future Work

This project established a robust implemented process for predicting solar power in Egypt. We successfully navigated the challenges of climate-induced variability through rigorous EDA, automated diagnostics, and iterative model testing.

Future Work: * **Irradiance Integration:** Incorporating direct solar irradiance measurements to replace weather proxies.

- **Hybrid Architectures:** Combining the temporal awareness of LSTMs with the stability of ExtraTrees.
- **Multi-year Expansion:** Extending the dataset to capture multi-year climate trends.